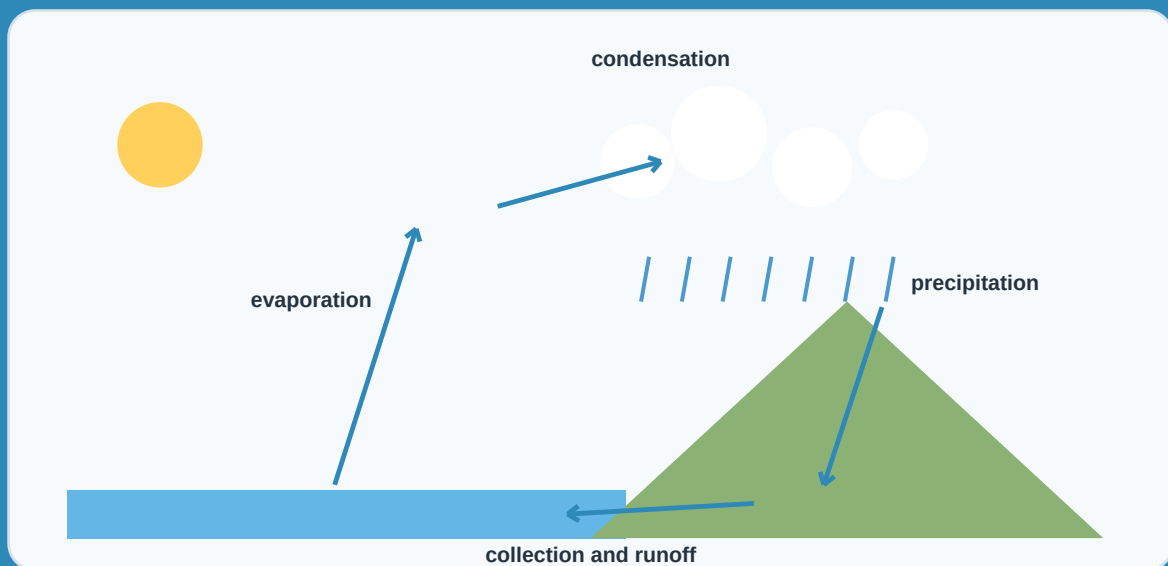


KATHAQUEST TEST CHAPTER

The Water Cycle: A Never-Ending Journey

Follow water as it moves through oceans, clouds, land and living things



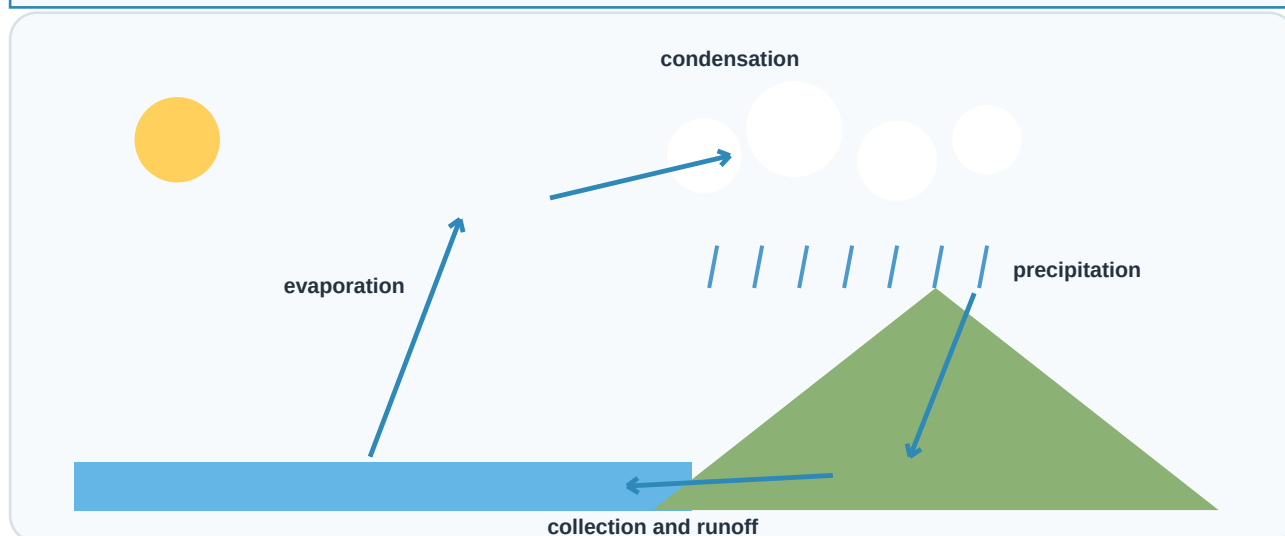
Recommended reading level: Ages 8-12

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What you will learn

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| 1 | Name the main stages of the water cycle. |
| 2 | Explain how the Sun powers evaporation. |
| 3 | Describe how clouds form and produce precipitation. |
| 4 | Identify ways water moves through land, plants and underground spaces. |

Key words: evaporation, condensation, precipitation, runoff, collection, groundwater, transpiration



1. Water is always moving

The water you drink today may once have been part of a cloud, a river or even an ancient ocean. Earth keeps reusing water through the **water cycle**. Water changes location and sometimes changes state, but the total amount of water on Earth stays nearly the same.

The Sun supplies most of the energy for this journey. Gravity pulls water down as rain, moves it downhill and helps it collect in lakes and oceans.

Imagine one water droplet travelling from the sea to a cloud, falling on a mountain and returning to the sea.

2. Evaporation and transpiration

When sunlight warms liquid water, some molecules move fast enough to escape into the air as invisible water vapour. This change from liquid to gas is called **evaporation**.

Plants also release water vapour through tiny openings in their leaves. This process is called **transpiration**. Together, evaporation and transpiration add moisture to the atmosphere.

Visual moment to look for: sunlight heating an ocean or puddle while vapour rises.

3. Condensation makes clouds

Warm, moist air rises and cools. Cooler air cannot hold as much water vapour, so vapour changes into tiny liquid droplets or ice crystals. This change is called **condensation**.

Billions of tiny droplets gather around small particles of dust, salt or smoke. From the ground, these groups of droplets look like clouds. Clouds are not made of water vapour alone; they contain real droplets and ice crystals.

You can see condensation when water droplets form on the outside of a cold glass.

4. Precipitation returns water

Cloud droplets collide and join together. When drops or ice crystals become too heavy to stay in the air, they fall as **precipitation**. Depending on temperature, precipitation may be rain, snow, sleet or hail.

Some water soaks into soil. Some flows across the ground as **runoff**. Streams and rivers carry runoff toward lakes and oceans.

Visual moment to look for: rain falling on hills and flowing into streams.

5. Collection and groundwater

Water collects in oceans, lakes, rivers, glaciers and wetlands. Some water moves downward through spaces in soil and rock. Underground water stored in these spaces is called **groundwater**.

Groundwater can slowly feed springs and rivers. People also reach it using wells. Eventually, much of this water returns to the ocean, where sunlight can begin the cycle again.

The water cycle has no true starting point. Any stage can lead into another.

Chapter summary

- The Sun provides energy for evaporation, while gravity moves water downward.
- Condensation forms cloud droplets when moist air cools.
- Precipitation includes rain, snow, sleet and hail.
- Water returns to rivers and oceans through runoff and groundwater.
- Plants join the cycle by releasing water vapour through transpiration.

Glossary

Word	Meaning
evaporation	The change of liquid water into water vapour.
condensation	The change of water vapour into liquid droplets.
precipitation	Water falling from clouds as rain, snow, sleet or hail.
runoff	Water flowing over the land surface.
groundwater	Water stored below the ground in soil and rock.
transpiration	The release of water vapour from plant leaves.

Quick check

1. What supplies most of the energy for the water cycle?
2. How is evaporation different from condensation?
3. What are four forms of precipitation?
4. Where can water collect after it falls?
5. How do plants add water to the atmosphere?

Testing note: This chapter is intentionally written with clear concepts, visual events and child-friendly language. It is suitable for testing chapter extraction, concept detection, media search, narration, quiz generation and revision-video workflows.